

LLM Models and Agents:

ChatGPT:

AI Overview

ChatGPT is an AI chatbot developed by OpenAI. It allows people to communicate with AI using normal, human-like language.

What is ChatGPT?

- Core Definition: ChatGPT was launched on November 30, 2022. GPT stands for Generative Pre-trained Transformer.
- ChatGPT is not open source.
- **Underlying Technology:** ChatGPT uses Large Language Models (LLMs) and deep learning to understand input and generate responses such as text, code, images, and audio.
- ChatGPT uses billions of adjustable internal weights called parameters that it fine-tunes during training to predict and generate human-like text.
- ChatGPT has used different LLMs over time. It is not always powered by one single model.
- **GPT-3.5** → Early ChatGPT model after its launch in 2022.
- **GPT-4** → Introduced stronger reasoning and accuracy.
- **GPT-4o** → Faster, multimodal model supporting text, image, and audio.
- **GPT-4.1 / GPT-4.1 mini** → Later-generation models, now retired from ChatGPT.
- **GPT-5 / GPT-5.1** → Later reasoning and general-purpose generations, also retired from normal ChatGPT use.
- **GPT-5.6 family** → Current ChatGPT generation:
 - **GPT-5.6 Luna** → default for Free and Go users.
 - **GPT-5.6 Sol** → used for higher-capability reasoning on eligible paid plans.
 - **GPT-5.6 Sol Pro** → higher-end Pro option.
- **GPT-6 Astra** → Available as **GPT-6 Pro** for eligible plans, including some ChatGPT Work/Codex access.

- **Training Method:** ChatGPT is trained using large amounts of data. It uses methods such as supervised learning and Reinforcement Learning from Human Feedback (RLHF) to improve its responses.

Key Features & Capabilities

- **Text & Conversation:** It can answer questions, explain topics, translate languages, summarize long documents, and write content.
- **Multimodal Tools:** It can work with different types of information such as text, images, files, and audio. It can also use tools such as web search, image generation, and data analysis.
- **Coding Support:** It can write, explain, debug, and improve code in different programming languages.

Limitations

- **Accuracy Issues:** ChatGPT can sometimes give wrong information or make up information. This is called a hallucination.
- **Not Human Understanding:** ChatGPT does not understand the world exactly like a human. It mainly learns patterns from data and uses those patterns to generate responses.

GEMINI:

Gemini is a family of natively multimodal AI models developed by Google DeepMind, designed to seamlessly understand and process text, images, audio, video, and code.

Core Overview

- **Developer:** Google
- Google Gemini model is not an open source model.
- **Nature:** Built from the ground up to be multimodal, meaning it processes different types of media simultaneously rather than stitching separate models together.
- **Interface:** Powers Google's primary AI assistant and app (accessible at the Gemini App), and integrates deeply into Google Workspace and Android.

Gemini Generation Timeline

- **Gemini 1.0 (December 2023)**
 - **Nano:** Built for light, on-device mobile tasks.
 - **Pro:** Designed for versatile mid-range text and multimodal tasks.
 - **Ultra:** Google's largest and most capable flagship model at launch.
- **Gemini 1.5 (2024)**
 - **1.5 Pro:** Introduced a massive 2-million-token context window for handling huge documents.
 - **1.5 Flash:** Optimized for high speed, low latency, and efficient distillation.
- **Gemini 2.0 (Late 2024 – Early 2025)**
 - **2.0 Flash / Flash-Lite:** Enhanced speed, native tool use, and real-time streaming support.
 - **Live API:** Enabled real-time multimodal voice and video interactions.
- **Gemini 2.5 (Mid 2025)**
 - **2.5 Pro:** Debuted Google's native chain-of-thought "thinking" capabilities for advanced reasoning.
 - **2.5 Flash / Flash-Lite:** Delivered high-performance price efficiency for high-volume developer tasks.
- **Gemini 3 and 3.8 Series (Late 2025 – 2026)**
 - **Gemini 3 / 3.1 Pro:** Advanced frontier reasoning and agentic workflows.
 - **Gemini 3.5, 3.6, 3.7, and 3.8 Flash:** Expanded speed, multimodal upgrades, and real-time live audio capabilities, with **3.8 Flash** serving as the newest workhorse

Key Model Tiers

- **Ultra:** Built for highly complex, heavy-duty reasoning and multi-step problem-solving.
- **Pro:** A balanced model optimized for scaling across a wide variety of advanced professional tasks.
- **Flash:** Designed for fast, high-frequency, everyday operations with low latency.
- **Nano:** The most efficient model tailored for on-device processing (such as on smartphones).

KIMI:

- Kimi is an AI chatbot and a family of LLMs developed by Moonshot AI.
- Kimi K3 is the latest major model mentioned in the file, released in July 2026.
- K3 has 2.8 trillion parameters and uses a Mixture-of-Experts (MoE) architecture.
- Kimi supports text, images, and video and has a 1-million-token context window in K3.
- Kimi also has agentic features, meaning it can perform multi-step tasks using tools and sub-agents.
- Important models include K1.5, K2, K2.5, K2.6, K2.7 Code, and K3.
- Kimi can be used through its web app and API, and some models have downloadable/open weights.
- The basic Kimi app is free with usage limits, with paid plans offering higher usage and additional features

CLAUDE:

- Claude is a family of Large Language Models (LLMs) developed by Anthropic.
- It was first released in March 2023.
- Claude is available as an AI chatbot and can also be used through an API.
- Claude models are based on Generative Pre-trained Transformers (GPTs).
- It is trained to predict the next word and then improved using RLHF and Constitutional AI.
- Claude has different model levels:
 - Haiku → smaller and faster
 - Sonnet → balanced
 - Opus → more powerful
 - Mythos → highest-level model with limited availability
- The file lists Claude Sonnet 5.5 and Claude Opus 5.5 as active models released in September 2026.
- Claude can help with writing, coding, research, documents, and other tasks.
- **Claude Code** is an AI coding agent that can perform coding tasks from the terminal using natural-language instructions.
- Claude Cowork is an agent for non-programmers that can work with files and perform multi-step tasks.
- Claude also supports features such as web search, computer use, and tool connections.
- Claude is mainly a cloud-based/proprietary AI, not an open-source LLM.

Model Levels and Generations:

Claude has different model generations and model levels.

- **Claude 1** → Early Claude generation
- **Claude 2** → Improved reasoning and longer context
- **Claude 3** → Haiku, Sonnet, Opus
- **Claude 3.5** → Improved coding and reasoning
- **Claude 3.7** → Added stronger reasoning capabilities
- **Claude 4** → Major improvement in coding and reasoning
- **Claude 4.5** → Haiku, Sonnet, Opus
- **Claude 4.6** → Newer generation with stronger coding/agent capabilities
- **Claude 5 / 5.x** → Newer generation listed in your uploaded source

Model levels:

Haiku → Fast and lightweight

Sonnet → Balanced speed + capability

Opus → More powerful for complex tasks

Claude models

- Claude Code (coding agent)
- Claude Cowork (file/computer agent)
- Claude in Chrome (browser agent)
- Claude Science (research/science agent)
- Claude Design (design agent)

These are not different Claude model versions; they are tools/agent products that use Claude models.

MANUS:

- Manus is an AI Agent, not just a normal chatbot.
- It was developed by Butterfly Effect.
- Manus was designed to perform tasks automatically, instead of only giving answers.
- It can take a goal, plan the steps, use different AI models/tools, and complete the task.
- Manus was launched in March 2025 as an invitation-only beta.
- The name Manus comes from Latin and means “hand.” It represents the idea of “mind + hand” — thinking and doing.

- Manus does not mainly train its own foundation LLM. It uses existing AI models as part of its agent system.

Versions:

Manus 1.0 → Earlier generation of the Manus agent.

Manus 1.5 → Improved speed, reliability, reasoning, and task completion. It had Manus 1.5 Lite.

Manus 1.6 → Improved planning, problem-solving, web development, and added mobile app development. It had:

- **1.6 Lite** → faster/lighter
- **1.6** → standard
- **1.6 Max** → more powerful for complex tasks

Manus 2.0 → Current major generation introduced in **September 2026**. It uses a new agent architecture called **Cascade** and adds things like Automations, Cloud Computer, and Manus Studio

MUSE:

- Muse is a personal AI Agent developed by Meta. It is designed to do tasks, not just answer questions.
- Muse is not just a chatbot. It is an AI agent.

- A normal chatbot mainly does:

You ask → AI answers.

- Muse is designed to do:

You give a goal → Muse plans → uses tools/apps → performs multiple steps → gives you the result.

Example:

Suppose you say:

"Find a good laptop under ₹60,000 and prepare a comparison."

Muse can:

- Understand your goal
- Search the web
- Collect laptop information
- Compare the options
- Create a document/table
- Return the result

What makes Muse interesting?

Muse has its own secure virtual computer + browser. This allows it to interact with websites rather than only generating text. Meta says it can also connect to things such as email, calendar and Instagram, and can even build tools when a required tool doesn't already exist.

Versions:

1. Muse Spark

- Muse Spark is a family of AI models developed by Meta.
- It is designed mainly for AI agents.
- It is more than a normal chatbot because it can reason, plan, use tools, and perform multi-step tasks.
- It can understand text, images, and other types of information.
- It can work with tools, computers, and MCP servers.
- It can also work with other AI agents, called subagents.

Muse Spark versions

Muse Spark 1.1

- Improved version of Spark.
- Better reasoning.
- Better tool use.
- Better computer use.
- Better coding.
- Better understanding of images and videos.
- Supports MCP servers.
- Can use subagents to divide a large task into smaller tasks.

Muse Spark 1.2

- Meta introduced Muse Code with Spark 1.2. Muse Code is a coding agent that can work on complex, multi-file code changes.
- Focused more on coding and software development.

- Can work with multiple files in a software project.
- Can understand existing code.
- Can make code changes.
- Can test the code and fix problems.

Muse Spark 1.3

- Newer version of the Spark family.
- Designed for long and complex agentic tasks.
- Spark 1.3 as being trained for **long-horizon agentic workflows**
- Better at maintaining context during long tasks.
- Can plan multiple steps.
- Can use tools and check the results.
- Better coding capabilities.
- Can correct its plan when something goes wrong.

2. Muse Glimmer

- Muse Glimmer is a separate model from the Spark family.
- It is also developed by Meta.
- It is designed especially for local AI agents.
- It is an open-weight model.
- It has 30 billion parameters (30B).
- It is released under the Apache 2.0 license.
- This makes it useful for developers who want to build and experiment with their own AI agents.
- Unlike cloud-only AI models, Glimmer is designed to be capable of running locally on consumer hardware.
- It can work with tools and function calling.
- It can be used for coding and agentic workflows.
- It can understand images/screenshots.
- It can work with local files and local tools.

Can I run Muse on my laptop?

- Yes, you can try Muse Glimmer on your laptop.
- Muse Glimmer is designed to support local AI agent workflows.
- It is a 30B (30-billion parameter) model.
- It is an open-weight model.
- It is designed to run on local/consumer hardware.
- Muse Spark is different. Spark is mainly used as a cloud/hosted model, so you normally don't download the full Spark model to your laptop.

If Your laptop has:

- 16 GB RAM
 - AMD graphics
 - 1 TB SSD
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- Because Glimmer is a 30B model, 16 GB RAM is quite limited for running it comfortably.
 - You may be able to try a quantized version, which uses less memory, but it may run slowly.

Command code:

- Command Code is an AI coding agent.
- It is designed to help developers write, understand, modify, and test code.
- You can give it instructions in normal English, instead of writing every coding step yourself.
- It can read your project files, understand the code, and make changes.
- It can also run commands and tests and help fix errors.
- The main idea is: you give the goal → Command Code plans → writes/changes code → tests it.

Versions

- **Command Code 0.x** → Early versions of the coding agent.
- **Command Code 1.0** → Major rebuilt version introduced in July 2026.
- **Command Code 1.x** → The product has continued receiving frequent updates and releases.
- **Latest version: 1.72.1** as currently listed on npm.

AI Tools Comparison:

AI Models & Agents Comparison							
A comprehensive feature comparison of major AI assistants and agents.							
Feature	ChatGPT	Gemini	Kimi	Claude	Manus	Muse	Command Code
Developed by	 OpenAI	 Google DeepMind	 Moonshot AI	 Anthropic	 Manus AI / Butterfly Effect	 Meta Superintelligence Labs	 Cohere
Type	AI Assistant	AI Assistant	AI Assistant + Agent	AI Assistant	AI Agent	Personal AI Agent	Coding / Enterprise AI
Main Speciality	General AI, reasoning, coding, research	Multimodal AI, long context, coding	Long context, coding, reasoning	Reasoning, coding, writing	Autonomous task execution	Personal task automation	Coding, reasoning, tool use
Important Versions	GPT-5.6 family	Gemini 3.1 family	Kimi K2 family	Claude Opus, Sonnet, Haiku	Manus 1.6, 1.6 Max, 2.0	Muse, Muse Spark	Command A family
Parameters	Not publicly disclosed	Not publicly disclosed	1T total / 328 active for K2	Not publicly disclosed	Not publicly disclosed	Not publicly disclosed	111B for Command A
Architecture	Proprietary	Proprietary	MoE for K2	Proprietary	Agent system using multiple models/tools	Agent system	Transformer / enterprise LLM
Free Version	Yes	Yes	Yes	Yes	Yes, limited	Availability varies	Mainly API / enterprise access
Paid Version	Yes	Yes	API / paid usage	Yes	Yes	Paid options may apply	Yes
Cloud / Local	Mainly Cloud	Mainly Cloud	Cloud + some self-hosting	Mainly Cloud	Mainly Cloud	Mainly Cloud	Cloud + self-hosting options
Open Source / Open Weights	No	No	Selected Kimi models	No	No	No	Open-weight options
Can Run on Normal Laptop?	Main model: No	Main model: No	Large K2: No	Main model: No	No	No	Not practical for normal laptops
Coding	Excellent	Excellent	Excellent	Excellent	Good	Good	Very strong
Reasoning	Strong	Strong	Strong	Strong	Uses agent planning	Agent planning	Strong
Long Documents	Yes	Very strong	Very strong	Very strong	Yes	Yes	Long context
Multimodal	Yes	Very strong	Yes, depending on model	Yes	Yes	Yes	Depends on model
Agentic Tasks	Yes	Yes	Yes	Yes	Core speciality	Core speciality	Yes
Tool Usage	Yes	Yes	Yes	Yes	Extensive	Extensive	Strong tool/function calling
Best Use	Learning, coding, research, general work	Google ecosystem, multimodal tasks	Long-context coding and reasoning	Coding, writing, reasoning	Autonomous complex tasks	Personal automation	Enterprise AI, coding and tool integration
Simple Description	AI that helps you think and work	Google's multimodal AI	Long-context and coding AI	Reasoning and coding AI	AI that performs tasks	Personal AI Agent	Enterprise AI for coding and tool use